

Types & Decisions

Homework: Variables, Types & if Statements

Name:

Date:

No computer needed! Show what you remember from class. Finish with a quick program you design yourself.

Part 1 — Matching

Write the letter of the correct meaning in the box next to each piece of code.

- A. Makes a variable whose value CAN change later.
- B. Makes a variable that is locked and cannot change.
- C. Says a variable must always hold text.
- D. Says a variable must always hold a number.
- E. Asks: are these two values equal? Gives true or false.
- F. Gives the remainder after dividing.
- G. Runs the code inside only when the answer is true.
- H. Lets you drop a variable into text, like `Hi \${name}`.

`let`

`const`

`: string`

`: number`

`===`

`%`

`if`

`` ${ } ``

Part 2 — True or False

Write T (true) or F (false) in each box.

- You can change the value of a const after you make it.
- A type annotation goes after the variable name, using a colon.
- Every comparison gives back a boolean (true or false).
- TypeScript only finds type mistakes after your program runs.
- An else block runs when the if question is false.

Part 3 — Predict the Output

Read each line carefully and write exactly what it would print.

Code:

```
console.log(7 + 3)
```

Prints:

Code:

```
console.log("7" + 3)
```

Prints:

Code:

```
console.log(9 % 2)
```

Prints:

Code:

```
console.log(5 > 10)
```

Prints:

Code:

```
let lives: number = 3
if (lives > 0) {
  console.log("Keep playing!")
} else {
  console.log("Game over")
}
```

Prints:

Part 4 — Spot the Error

Each snippet has ONE mistake. Circle it or describe it, then write the fix.

Snippet 1:

```
const city = "San Jose"
city = "Oakland"
```

What's wrong, and how do you fix it?

Snippet 2:

```
let age: number = "twelve"
```

What's wrong, and how do you fix it?

Part 5 — Short Answer

1. Explain the difference between `let` and `const` in your own words.

2. Why is it helpful for TypeScript to check your types before the program runs?

Part 6 — Design Your Own Program

Plan a small program on paper. It must use a typed variable and an `if/else`.

What does your program do?

Write your code here (it's okay if it isn't perfect!):